

Joint Before Separation in a Primitive Möbius Tail: Both-New Annuli, Sharp Symmetric Conductor Ledgers, and Cutoff Stability

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July 2026

Abstract

Let $P_{m,W}$ be the calibrated residual prefix in a primitive opened- d row and let $R \leq S < T$. TPC-23 closes a centered selected square below the Poisson length, while TPC-24–25 close the two mixed increments obtained by replacing one prefix at S by the new shell $L_{m;S,T} = P_{m,T} - P_{m,S}$. The remaining component is the both-new square $L_{m;S,T}L_{n;S,T}$. We treat a stronger object: the whole quadratic cutoff increment

$$P_{m,T}P_{n,T} - P_{m,S}P_{n,S}.$$

The decisive step is exact and precedes every smooth or spectral separation. Writing

$$P_{m,W}(j) = \sum_{\substack{u \leq W \\ u|mj+h}} b_R(u) - \delta_H(m),$$

the cutoff difference is one raw annular joint divisor lattice, together with two explicit row-drift terms. The annulus contains both mixed orientations and the both-new square, but no surviving single-indicator Poisson channel.

For the new shell-shell physical zero kernel we prove, uniformly for off-diagonal rows, $S = X^{s+o(1)}$, every $0 < \kappa < s$, and every fixed $A > 0$,

$$K_{S,T}^{\square,0}(m, n) \ll (\log X)^{-A} + X^{-s+\kappa+o(1)}.$$

The small shared factors are controlled by cancellation between the two endpoints of a complete Möbius shell; the large shared factors divide the nonzero determinant $m - n$. TPC-25's actual prime-Möbius row provenance supplies $O(Q^2)$ absolute row-pair mass, so the complete annular zero frequency has arbitrary logarithmic saving for a general fixed nonzero-mean physical test.

At nonzero frequency we prove a symmetric exact-conductor box theorem. If $Q = X^{\beta+o(1)}$, $J = X^{1-\beta+o(1)}$, and $T = X^{t+o(1)}$, the exact minimax saving over the relaxed conductor rectangle is denoted by $M_\beta(t)$. In the Poisson-beyond region $t > 1 - \beta$,

$$M_\beta(t) = \begin{cases} \frac{3(1-\beta)-2t}{4}, & 1-\beta < t \leq \beta/2, \\ \min \left\{ \frac{\beta+1-4t}{2}, \frac{3-\beta-6t}{4} \right\}, & t \geq \beta/2. \end{cases}$$

Thus the annular conductor argument has a nonempty beyond-Poisson region for every $3/5 < \beta < 1$; the former face $\beta < 2/3$ came from a nonsharp low-low estimate. Under the

inherited primitive, no-wrap, row-mask, and fixed-power interfaces, and with $M_\beta(t) > 0$ by a fixed margin, the actual physical cutoff increment is

$$\mathcal{Q}_T^{\text{phys}} - \mathcal{Q}_S^{\text{phys}} \ll_B XQ(\log X)^{-B}$$

for every fixed $B > 0$. A strict sample with $S < J < T$ has saving $1/100$. A second, narrower sample with $\beta > 2/3$ is compatible with the published opened- d source geometry and has saving $33/8000$, but proves cutoff stability only: its base square is not imported as closed. If an independent matching calibrated S -cutoff estimate is supplied, the result closes the selected calibrated square through $T > J$. Such a calibrated base estimate is not deduced here from the earlier centered-square results. The paper does not estimate the ultra-long $(T, U]$ complement, prove positivity, produce a Hardy–Littlewood main term, imply twin primes, or breach the general sieve parity barrier.

Keywords: primitive Möbius tail; both-new shell; Poisson boundary; exact conductor; cutoff stability; sieve parity.

MSC 2020: Primary 11N35, 11N37; Secondary 11L07, 11L40.

1 Introduction

The primitive Möbius-tail program of TPC-18–25 reorganizes a fixed-shift prime-pair residual into dyadic opened rows

$$m = \ell d, \quad \ell \asymp L \text{ prime}, \quad d \asymp D, \quad Q = LD, \quad QJ \asymp X.$$

TPC-18 isolates the generic off-diagonal row correlation, TPC-19–20 derive its exact divisor and joint-Poisson normal forms, TPC-21 closes the Ramanujan mean, and TPC-22–23 control a selected square of divisor coordinates by exact-conductor large-sieve estimates [3–8]. TPC-24 then lets one new divisor leg pass beyond the old Poisson length, and TPC-25 removes the remaining projective-mass hypothesis for the actual opened- d packet by extracting physical frequency zero before smooth row–orbit separation [9, 10].

The object left between those results is elementary to state. For a calibrated residual prefix $P_{m,W}$, put

$$L_{m;S,T} = P_{m,T} - P_{m,S}, \quad R \leq S < T. \quad (1)$$

The quadratic cutoff difference expands as

$$P_{m,T}P_{n,T} - P_{m,S}P_{n,S} = P_{m,S}L_{n;S,T} + L_{m;S,T}P_{n,S} + L_{m;S,T}L_{n;S,T}. \quad (2)$$

The first two pieces are the one-sided increments of TPC-24–25. The third is the both-new square. Estimating the three terms separately, however, hides a more useful exact identity. The present paper first recombines the whole left side of (2), before either divisor leg is centered, Mellin-separated, or sent through a Fourier channel.

1.1 Joint before separation

Let

$$a(u) = -\mu(u) \log u, \quad b_R(u) = a(u) - \lambda'_R(u), \quad \lambda'_R(u) = 0 \quad (u > R).$$

The calibrated prefix identity of TPC-24 is

$$P_{m,W}(j) = \sum_{\substack{u \leq W \\ u \mid mj+h}} b_R(u) - \delta_H(m). \quad (3)$$

Because $S \geq R$, every new coefficient is $b_R(u) = a(u)$. Direct subtraction gives

$$\begin{aligned} & P_{m,T}(j)P_{n,T}(j) - P_{m,S}(j)P_{n,S}(j) \\ &= \sum_{\substack{u,v \leq T \\ \max\{u,v\} > S}} b_R(u)b_R(v) \mathbf{1}_{u \mid mj+h} \mathbf{1}_{v \mid nj+h} \\ & \quad - \delta_H(m)L_{n,S,T}(j) - \delta_H(n)L_{m,S,T}(j). \end{aligned} \quad (4)$$

The first term is one raw joint lattice on the annulus

$$\{(u, v) : u, v \leq T, \max\{u, v\} > S\}.$$

It contains both mixed orientations and the both-new square. More importantly, it contains no single-indicator Poisson channel. The two remaining terms are explicit row drifts, smaller by L^{-1} on the physical packet. This order of operations is the structural reason that the nonzero spectrum may be treated with a symmetric conductor box even when $T > J$.

1.2 The new zero-mode theorem

The mixed parts of the annulus inherit the pointwise shell theorem of TPC-24. For the new shell-shell corner, the zero kernel is

$$K_{S,T}^{\square,0}(m, n) = \sum_{\substack{S < u, v \leq T \\ (u, mH) = (v, nH) = 1}} \frac{a(u)a(v)}{[u, v]} \mathbf{1}_{(u,v) \mid m-n}. \quad (5)$$

Writing $g = (u, v)$, $u = gc$, and $v = gd$ makes (g, c, d) pairwise coprime on the squarefree support. If $g \leq SX^{-\kappa}$, the innermost d -sum is the difference of two complete Möbius calibrations at T/g and S/g ; their common main constant cancels. If $g > SX^{-\kappa}$, the off-diagonal condition gives $g \mid m - n \neq 0$. We obtain

$$\boxed{K_{S,T}^{\square,0}(m, n) \ll_{A,\kappa} (\log X)^{-A} + X^{-s+\kappa+o(1)}} \quad (6)$$

for $S = X^{s+o(1)}$, every fixed $A > 0$, and every $0 < \kappa < s$. TPC-25 proves that the actual prime-Möbius row coefficients have $O(Q^2)$ absolute pair mass before smooth separation. Consequently (6) closes the physical annular zero mode without a mean-zero test, a row-balance hypothesis, or a new projective norm.

1.3 A sharp symmetric conductor ledger

At nonzero additive frequency the annular restriction only reduces the nonnegative conductor energy. Enlarge it to $u, v \leq T$, resolve the two exact coordinate conductors, and combine the restricted-frequency norm route with the induced-Gauss route. Write

$$Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad T = X^{t+o(1)}. \quad (7)$$

For a denominator cell $Y = X^{y+o(1)}$ and conductor cells $F_i = X^{f_i+o(1)}$, put

$$k_i = \max\{f_i, \beta - f_i\}.$$

The two saving exponents are

$$\eta_A = \frac{\beta + 1 - y - k_1 - k_2}{2}, \quad \eta_B = 1 - f_1 - f_2 - \frac{k_1 + k_2}{2}. \quad (8)$$

Here $y \leq 2t$ and $0 \leq f_i \leq t$. We compute the exact relaxed- box minimax

$$M_\beta(t) = \inf_{0 \leq f_1, f_2 \leq t} \max\{\eta_A(2t, f_1, f_2), \eta_B(f_1, f_2)\}. \quad (9)$$

For $1/2 \leq \beta < 1$, it is the continuous piecewise-linear function

$$M_\beta(t) = \begin{cases} 1 - \beta - t, & 0 \leq t \leq (1 - \beta)/2, \\ \frac{3 - 3\beta - 2t}{4}, & (1 - \beta)/2 \leq t \leq \beta/2, \\ \frac{3 - \beta - 6t}{4}, & \beta/2 \leq t \leq (3\beta - 1)/2, \\ \frac{\beta + 1 - 4t}{2}, & t \geq (3\beta - 1)/2. \end{cases} \quad (10)$$

The previous low-low lower bound $1 - 3\beta/2$ is not an exact minimax face when $t < \beta/2$. In the region $t > 1 - \beta$, some positive interval survives for every

$$\frac{3}{5} < \beta < 1. \quad (11)$$

This is a statement about the annular conductor problem, not about the base square or the complete residual. The equality witnesses in the relaxed box need not all occur arithmetically; the minimax remains a valid sufficient bound because the arithmetic cells form a subset.

1.4 Main consequence and exact scope

Let $\mathcal{Q}_W^{\text{phys}}$ be the selected physical quadratic packet formed with $P_{m,W}P_{n,W}$, the same row family, the same smooth orbit weight, and the same divisor-independent off-diagonal mask at both cutoffs. Under the primitive opened-row interfaces and fixed positive margins, the main theorem gives

$$\boxed{\mathcal{Q}_T^{\text{phys}} - \mathcal{Q}_S^{\text{phys}} \ll_{B, \mathcal{D}} XQ(\log X)^{-B}} \quad (12)$$

for every fixed $B > 0$, whenever $M_\beta(t) > 0$. Here $\mathcal{D}$ denotes the fixed shift, residue, smoothness, no-wrap, and source data. The smooth Fourier tail is proved negligible for the actual provenance rather than retained as an independent packet hypothesis.

If an independent matching calibrated S -cutoff estimate is available, (12) closes the selected calibrated square through T . This implication is stated separately: cutoff stability alone does not estimate its initial value, and the scalar reassembly from the earlier centered base packet is not proved here.

The first rational sample satisfies $S < J < T$ and has conductor saving $1/100$. A second sample has $\beta = 267/400 > 2/3$ and saving $33/8000$. It lies inside the published opened- d row geometry, but both annular endpoints are already beyond J , so it establishes only beyond-face cutoff stability.

Nothing in the paper estimates $L_{:,T,U}$, the ultra-long complement identified in TPC-23. That complement consists of growing two-point affine Möbius fibers; known fixed-form logarithmic

results do not supply the required polynomially uniform rate [2, 8]. Global endpoint reassembly, positivity, the prime-pair main term, Hardy–Littlewood, twin primes, and a general breach of the parity barrier therefore remain outside the conclusion.

2 Physical rows and exact annular recombination

We fix the packet carefully because the cutoff identity is valid only when the two quadratic forms use the same physical rows, mask, and orbit weight.

2.1 Primitive opened rows

Fix $h \in \mathbb{Z} \setminus \{0\}$, and put

$$H = \text{rad } |h|, \quad H_0 = |h|. \quad (13)$$

Choose

$$R = X^{1/2-\delta+o(1)}, \quad 0 < \delta < \frac{1}{2}, \quad R \leq S < T \leq X^{C_T}, \quad (14)$$

where all exponent inequalities used below have fixed margins. In one opened- d slice the rows are indexed by

$$\alpha = (\ell, d), \quad m_\alpha = \ell d \asymp Q = LD, \quad (15)$$

with

$$\ell \in [L/2, 2L] \text{ prime}, \quad d \in [D, 2D] \cap [1, R^{1/2}], \quad d \text{ squarefree}, \quad (d, H) = 1. \quad (16)$$

We retain

$$QJ \asymp X, \quad L > \max\{2R, 2|h|\}, \quad L \geq X^{\lambda_0} \quad (17)$$

for a fixed $\lambda_0 > 0$. Target primitivity gives

$$(w, m_\alpha H) = 1 \quad \text{whenever } w \mid m_\alpha j + h. \quad (18)$$

The two row coefficients have the actual provenance

$$\gamma_\alpha^{(i)} = \mu(d)(\log \ell)\omega_D^{(i)}(d)\psi_L^{(i)}(\ell/L)\zeta_\alpha^{(i)}, \quad |\zeta_\alpha^{(i)}| \leq C, \quad i = 1, 2, \quad (19)$$

where the smooth dyadic and fixed residue factors range over bounded families. TPC-25 proves, uniformly in such data,

$$\sum_\alpha |\gamma_\alpha^{(i)}| \ll Q, \quad (20)$$

$$\|\gamma^{(i)}\|_\infty \ll \log(2L), \quad \|\gamma^{(i)}\|_2^2 \ll Q \log(2L). \quad (21)$$

Let $\mathbf{m}(\alpha, \gamma)$ be the actual generic row-pair mask. We use

$$|\mathbf{m}(\alpha, \gamma)| \leq 1, \quad \mathbf{m}(\alpha, \gamma) = 0 \quad \text{if } m_\alpha = m_\gamma, \quad (22)$$

and assume that the mask is independent of every divisor and conductor variable. The second condition is structural: the large-shared-factor zero estimate uses $m_\alpha - m_\gamma \neq 0$.

Let $\Xi_{\alpha,\gamma}$ be a uniformly bounded real periodic factor of fixed period H_0 , and let $\mathcal{W}_{\alpha,\gamma}$ range over the factorable row-orbit smooth family of TPC-25. In particular, before smooth separation,

$$\sup_{\alpha,\gamma} \left| \int_{\mathbb{R}} \mathcal{W}_{\alpha,\gamma}(t) dt \right| \ll 1, \quad (23)$$

and its continuous Mellin representation has finite weighted variation of every fixed order. These conditions are automatic for the fixed dyadic physical cutoffs under consideration; they are not asserted for an arbitrary row-dependent smooth family.

Throughout the paper, $\mathscr{D}$ denotes the fixed data

$$h, H_0, C_T, \delta, \lambda_0, \varepsilon_0,$$

together with all fixed positive exponent margins and the bounded smooth, dyadic, sign, and residue families above. Implied constants may depend on $\mathscr{D}$, but not on X or on a row in the retained packet.

The annular theorem below uses only this physical-shaped interface. When we identify a sample with the published Maynard source profile used in TPC-18, we additionally verify the source relations

$$J = K/D, \quad D_0 < D \leq V, \quad K > 2V.$$

Those extra relations certify provenance of that sample; they are not silently inserted into the abstract annular argument.

2.2 Calibrated prefixes

Put

$$a(w) = -\mu(w) \log w, \quad (24)$$

and define the finite Ramanujan coefficient

$$\lambda'_R(w) = w \sum_{b \leq R/w} \frac{\mu(wb)\mu(b)}{\varphi(wb)} \quad (w \leq R), \quad \lambda'_R(w) = 0 \quad (w > R). \quad (25)$$

Set

$$b_R(w) = a(w) - \lambda'_R(w). \quad (26)$$

All three coefficients vanish off squarefree integers. For a row $m = \ell d$, let

$$\delta_H(m) = \frac{H}{\varphi(H)} \frac{d}{\varphi(d)(\ell-1)} \ll_{\varepsilon,h} \frac{X^\varepsilon}{L}. \quad (27)$$

The exact calibrated prefix from TPC-24 is

$$P_{m,W}(j) = \sum_{\substack{w \leq W \\ w|mj+h}} b_R(w) - \delta_H(m), \quad R \leq W. \quad (28)$$

Consequently, for $R \leq S < T$,

$$L_{m;S,T}(j) := P_{m,T}(j) - P_{m,S}(j) = \sum_{\substack{S < w \leq T \\ w|mj+h}} a(w). \quad (29)$$

The restriction $S \geq R$ is used here: it ensures $b_R(w) = a(w)$ on the entire new shell. For later use define the raw prefix

$$A_{m,W}(j) := \sum_{\substack{w \leq W \\ w|mj+h}} b_R(w), \quad P_{m,W} = A_{m,W} - \delta_H(m). \quad (30)$$

Proposition 2.1 (Exact annular prefix identity). *For $R \leq S < T$, pointwise in m, n, j ,*

$$\begin{aligned} & P_{m,T}(j)P_{n,T}(j) - P_{m,S}(j)P_{n,S}(j) \\ &= \sum_{\substack{u,v \leq T \\ \max\{u,v\} > S}} b_R(u)b_R(v)\mathbf{1}_{u|mj+h}\mathbf{1}_{v|nj+h} \\ & \quad - \delta_H(m)L_{n,S,T}(j) - \delta_H(n)L_{m,S,T}(j). \end{aligned} \quad (31)$$

Equivalently, the raw annulus is the disjoint sum of the b_R - a mixed rectangle, its transpose, and the a - a both-new square.

Proof. By (30), the difference of the two products is

$$\begin{aligned} & A_{m,T}A_{n,T} - A_{m,S}A_{n,S} - \delta_H(m)(A_{n,T} - A_{n,S}) \\ & \quad - \delta_H(n)(A_{m,T} - A_{m,S}). \end{aligned}$$

The $\delta_H(m)\delta_H(n)$ terms cancel exactly. The first difference is the raw joint sum over $u, v \leq T$ with $\max\{u, v\} > S$, while each one-row difference is (29). This proves the formula. $\square$

2.3 The physical quadratic packet

Using the same physical weight at both cutoffs, define

$$\begin{aligned} \mathcal{Q}_W^{\text{phys}} &:= \sum_{\alpha, \gamma} \gamma_\alpha^{(1)} \gamma_\gamma^{(2)} \mathbf{m}(\alpha, \gamma) \\ & \quad \times \sum_{\substack{j \in \mathbb{Z} \\ (j, H)=1}} \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J) P_{m_\alpha, W}(j) P_{m_\gamma, W}(j). \end{aligned} \quad (32)$$

The natural row-orbit scale is

$$JQ^2 \asymp XQ. \quad (33)$$

Lemma 2.2 (Drift contribution). *The two drift terms in (31) contribute to $\mathcal{Q}_T^{\text{phys}} - \mathcal{Q}_S^{\text{phys}}$ at most*

$$\ll_{\varepsilon, \mathcal{D}} JQ^2 L^{-1} X^\varepsilon. \quad (34)$$

Proof. On the fixed orbit support,

$$|L_{m,S,T}(j)| \leq \sum_{w|mj+h} |a(w)| \ll_\varepsilon X^\varepsilon.$$

There are $O_{\mathcal{D}}(J)$ orbit points. Apply (27), delete the bounded mask, and use (20) on both row sides. $\square$

Remark 2.3 (Cutoff-independent physical weights). The identity compares two cutoffs inside one fixed physical packet. If the smooth partition, residue factor, row family, or mask is changed with W , an additional weight-difference term appears and is not covered by [proposition 2.1](#).

3 The both-new shell at physical frequency zero

The nonzero conductor argument has no information at additive frequency zero. We therefore isolate that frequency before introducing characters or separating the physical row-orbit weight.

3.1 Uniform endpoint calibration

For $M, Y \geq 1$, set

$$A_M(Y) = \sum_{\substack{w \leq Y \\ (w, M)=1}} \frac{a(w)}{w}, \quad B_M(Y) = \sum_{\substack{w \leq Y \\ (w, M)=1}} \frac{\mu(w)}{w}. \quad (35)$$

We use the following uniform form proved in TPC-24 from the classical zero-free region for $\zeta(s)$ [1, 9].

Lemma 3.1 (Uniform shell calibration). *For every fixed $A, C > 0$, uniformly for $Y \geq 3$ and $M \leq Y^C$,*

$$A_M(Y) = \frac{M}{\varphi(M)} + O_{A,C} \left(\frac{M}{\varphi(M)} (\log Y)^{-A} \right), \quad (36)$$

$$B_M(Y) \ll_{A,C} \frac{M}{\varphi(M)} (\log Y)^{-A}. \quad (37)$$

Consequently, if $Y_2 \geq Y_1 \geq X^\kappa$ and $M \leq X^{C_0}$, then both endpoint differences

$$A_M(Y_2) - A_M(Y_1), \quad B_M(Y_2) - B_M(Y_1) \quad (38)$$

are $O_{A,C_0,\kappa}((\log X)^{-A})$ after increasing the input calibration exponent by a fixed amount.

The first main constant $M/\varphi(M)$ is independent of Y , so it cancels only after the complete interval is retained. This is not a pointwise fixed-modulus statement.

3.2 Exact gcd reparametrization

For an off-diagonal row pair $m \neq n$, put

$$\Delta = m - n \neq 0. \quad (39)$$

The raw Poisson zero frequency of the both-new joint shell is governed by

$$K_{S,T}^{\square,0}(m,n) = \sum_{\substack{S < u,v \leq T \\ (u,mH)=(v,nH)=1}} \frac{a(u)a(v)}{[u,v]} \mathbf{1}_{(u,v)|\Delta}. \quad (40)$$

All effective variables are squarefree.

Lemma 3.2 (Both-new gcd normal form). *The kernel in (40) equals*

$$K_{S,T}^{\square,0}(m,n) = \sum_{\substack{g|\Delta \\ g \text{ squarefree} \\ (g,mnH)=1}} \frac{1}{g} \sum_{\substack{S/g < c \leq T/g \\ (c,gmH)=1}} \frac{a(gc)}{c} \\ \times \sum_{\substack{S/g < d \leq T/g \\ (d,gcnH)=1}} \frac{a(gd)}{d}. \quad (41)$$

Proof. For one effective pair write

$$g = (u, v), \quad u = gc, \quad v = gd.$$

Squarefree support and exactness of the gcd make g, c, d pairwise coprime. Moreover,

$$[u, v] = g c d, \quad \frac{1}{[u, v]} = \frac{1}{g c d}.$$

Compatibility is precisely $g \mid \Delta$. The primitive conditions become $(g, mnH) = 1$, $(c, gmH) = 1$, and $(d, gcnH) = 1$. Substitution gives the formula. $\square$

3.3 Pointwise cancellation

Theorem 3.3 (Both-new pointwise zero theorem). *Suppose*

$$S = X^{s+o(1)}, \quad T \leq X^{C_T}, \quad s, C_T > 0. \quad (42)$$

For every fixed $A > 0$ and $0 < \kappa < s$, uniformly over the retained off-diagonal row pairs,

$$\boxed{K_{S,T}^{\square,0}(m,n) \ll_{A,\kappa,C_T,h} (\log X)^{-A} + X^{-s+\kappa+o(1)}}. \quad (43)$$

Proof. In (41) split the shared factor at

$$G_0 = SX^{-\kappa}. \quad (44)$$

Suppose first that $g \leq G_0$. Then $S/g \geq X^\kappa$. Fix g, c , put

$$M = gcnH, \quad (45)$$

and note that $M \leq X^{C_0}$ for a fixed C_0 , uniformly on the physical fixed-power ranges. On the effective coprime support,

$$a(gd) = \mu(g)a(d) - \mu(g)(\log g)\mu(d). \quad (46)$$

Hence the innermost sum in (41) is exactly

$$\mu(g)\{A_M(T/g) - A_M(S/g)\} \\ - \mu(g)(\log g)\{B_M(T/g) - B_M(S/g)\}. \quad (47)$$

By lemma 3.1, this has arbitrary logarithmic decay. The outer cofactor mass obeys

$$\sum_{c \leq T/g} \frac{|a(gc)|}{c} \ll_{C_T} (\log(2X))^2, \quad (48)$$

and

$$\sum_{g|\Delta} \frac{1}{g} \ll \log \log(3|\Delta|).$$

Increasing the calibration exponent proves

$$K_{S,T}^{\square,0,g \leq G_0}(m,n) \ll_A (\log X)^{-A}. \quad (49)$$

For $g > G_0$, take absolute values in both cofactor sums. Each is bounded by a fixed power of $\log X$, and the nonzero determinant gives

$$\sum_{\substack{g|\Delta \\ g > G_0}} \frac{1}{g} \leq \frac{\tau(|\Delta|)}{G_0} = X^{-s+\kappa+o(1)}. \quad (50)$$

Combining (49) and (50) proves the theorem. $\square$

Remark 3.4 (Why the principal row mode disappears here). TPC-25 exhibits a finite shell with a nonzero constant $g = 1$ row kernel. In the complete both-new interval, the $g = 1$ coefficient is an endpoint difference of the form (47); its main constant cancels. Thus the physical proof does not require projection away from the constant row vector. This conclusion depends on retaining the complete shell, not on a fixed-modulus calibration assertion.

3.4 The complete raw annulus

Define the two mixed kernels

$$K_{S,T}^{ba,0}(m,n) := \sum_{\substack{u \leq S, \ S < v \leq T \\ (u,mH)=(v,nH)=1}} \frac{b_R(u)a(v)}{[u,v]} \mathbf{1}_{(u,v)|\Delta}, \quad (51)$$

$$K_{S,T}^{ab,0}(m,n) := K_{S,T}^{ba,0}(n,m). \quad (52)$$

The raw annular joint zero kernel is

$$K_{S,T}^{\text{ann},0}(m,n) = K_{S,T}^{ba,0}(m,n) + K_{S,T}^{ab,0}(m,n) + K_{S,T}^{\square,0}(m,n). \quad (53)$$

TPC-24 proves the analogue of (43) for $K^{ba,0}$, and transposition gives it for $K^{ab,0}$. We therefore obtain:

Corollary 3.5 (Annular raw zero kernel). *Under the hypotheses of [theorem 3.3](#),*

$$\boxed{K_{S,T}^{\text{ann},0}(m,n) \ll_{A,\kappa,C_T,h} (\log X)^{-A} + X^{-s+\kappa+o(1)}}. \quad (54)$$

The two single-density drift terms from [proposition 2.1](#) are not part of (53); they remain explicit and are controlled by [lemma 2.2](#).

4 Physical zero before separation

We now insert the arithmetic kernel into the actual opened- d packet. The order matters: physical additive frequency zero is taken while the row-dependent orbit weight is still one bounded smooth function.

4.1 The fixed local factor

Because every effective joint modulus is coprime to $H_0 = |h|$, the fixed periodic factor may be split into finitely many residue cells. For one row pair define its zero-frequency local average by

$$\vartheta_{H_0}(\alpha, \gamma) = \frac{1}{H_0} \sum_{\substack{r \bmod H_0 \\ (r, H)=1}} \Xi_{\alpha, \gamma}(r). \quad (55)$$

It is uniformly bounded. The zero Fourier functional of the smooth orbit amplitude is

$$\widehat{\mathcal{W}}_{\alpha, \gamma}(0) = \int_{\mathbb{R}} \mathcal{W}_{\alpha, \gamma}(t) dt, \quad \sup_{\alpha, \gamma} |\widehat{\mathcal{W}}_{\alpha, \gamma}(0)| \ll 1. \quad (56)$$

Thus, for one row pair, the raw annular zero contribution has the explicit form

$$J \vartheta_{H_0}(\alpha, \gamma) \widehat{\mathcal{W}}_{\alpha, \gamma}(0) K_{S, T}^{\text{ann}, 0}(m_\alpha, m_\gamma). \quad (57)$$

All factors preceding the kernel are uniformly bounded. When $\Xi \equiv 1$, the local average reduces to the familiar fixed primitive density; the proof does not require that special case.

4.2 The provenance-weighted theorem

Let $\mathcal{Z}_{S, T}^{\text{ann}, 0}$ denote the physical zero-frequency contribution of the raw annular joint lattice in (31), including the local factor (55), but excluding the two separately displayed drift terms.

Theorem 4.1 (Physical annular zero theorem). *Retain the physical provenance (19), the off-diagonal mask (22), and the hypotheses of theorem 3.3. Then, for every fixed $A > 0$ and $0 < \kappa < s$,*

$$|\mathcal{Z}_{S, T}^{\text{ann}, 0}| \ll_{A, \kappa, \mathcal{D}} J Q^2 \left\{ (\log X)^{-A} + X^{-s+\kappa+o(1)} \right\}. \quad (58)$$

After the zero contribution is combined with the two separately bounded full drift contributions,

$$|\mathcal{Z}_{S, T}^{\text{cutoff}, 0}| \ll J Q^2 \left\{ (\log X)^{-A} + X^{-s+\kappa+o(1)} + L^{-1} X^\varepsilon \right\}. \quad (59)$$

Proof. By corollary 3.5, each retained row pair has raw zero kernel bounded by the braces in (58). Delete the bounded mask and the bounded local and smooth zero factors. The actual row provenance gives

$$\sum_{\alpha, \gamma} |\gamma_\alpha^{(1)} \gamma_\gamma^{(2)} \mathbf{m}(\alpha, \gamma)| \leq \left(\sum_{\alpha} |\gamma_\alpha^{(1)}| \right) \left(\sum_{\gamma} |\gamma_\gamma^{(2)}| \right) \ll Q^2$$

by (20). Multiplication by the orbit length J proves the first assertion. The second follows from lemma 2.2. $\square$

Corollary 4.2 (Arbitrary logarithmic zero saving). *Assume $s > 0$ and $\lambda_0 > 0$ with fixed margins. For every fixed $B > 0$, choose $\kappa < s$, the auxiliary $\varepsilon < \lambda_0$, and then the calibration exponent A . Since $JQ^2 \asymp XQ$,*

$$\boxed{\mathcal{Z}_{S,T}^{\text{cutoff},0} \ll_{B,\mathscr{D}} XQ(\log X)^{-B}.} \quad (60)$$

No mean-zero condition is imposed on the fixed physical test. The conclusion is a provenance theorem for the actual opened- d packet; it is not inferred from the abstract soft row-energy bound alone.

4.3 Transfer to the nonzero interface

Only after the zero frequency is removed do we separate the smooth row-orbit factors. First split the fixed periodic multiplier into its $O_h(1)$ residue cells:

$$\Xi_{\alpha,\gamma}(j) = \sum_{\varrho \bmod H_0} \Xi_{\alpha,\gamma}(\varrho) \mathbf{1}_{j \equiv \varrho \pmod{H_0}}. \quad (61)$$

On a fixed cell, the bounded row-pair coefficient $\Xi_{\alpha,\gamma}(\varrho)$ is absorbed into the generic mask. It remains divisor-independent, bounded, and supported off the row diagonal. The extra congruence changes the Poisson period by only the fixed factor H_0 , and hence changes none of the conductor exponents. The following is the precise form of the TPC-25 transfer lemma needed below.

Lemma 4.3 (Continuous physical transfer). *For every fixed residue cell $\varrho \bmod H_0$, the physical coefficient-weight product has the continuous projective representation*

$$\gamma_\alpha^{(1)} \gamma_\gamma^{(2)} \mathcal{W}_{\alpha,\gamma}(j/J) = \int_{\mathbb{R}^4} x_\tau(\alpha) y_\tau(\gamma) \Phi_\tau(j/J) d\nu(\tau). \quad (62)$$

For every fixed integer $M \geq 0$, it may be chosen so that

$$\int_{\mathbb{R}^4} (1 + |\tau|)^M \|x_\tau\|_2 \|y_\tau\|_2 d|\nu|(\tau) \ll_M Q \log(2L), \quad (63)$$

$$\int_{\mathbb{R}^4} (1 + |\tau|)^M \|x_\tau\|_1 \|y_\tau\|_1 d|\nu|(\tau) \ll_M Q^2. \quad (64)$$

The row scales, target primitivity, and divisor-independent generic mask are unchanged. Every fixed orbit seminorm grows at most polynomially in τ , and summing the $O_h(1)$ residue cells followed by dominated convergence restores the original physical packet after componentwise estimates.

Proof. Mellin inversion separates each fixed smooth factor depending on $m_\alpha j/X$, on the normalized opened-divisor argument $(d/D)(j/J)$, or on the corresponding second-row variables. Unit-modulus Mellin phases do not change the row norms (21). Schwartz decay gives every weighted moment of the Mellin measure. The ℓ^1 statement is (20) before integration. Differentiating the orbit phases produces a polynomial in τ , which is absorbed by a higher Schwartz moment. The generic mask depends only on the row pair and therefore remains outside the integral. These are precisely the transfer steps established for the same physical provenance in TPC-25 [10]. $\square$

Lemma 4.4 (The physical Fourier tail is automatic). *Fix $0 < \omega < \varepsilon_0$, where the no-wrap margin below is $J \geq 4HX^{\varepsilon_0}$. For every prescribed $A > 0$, repeated integration by parts in the fixed smooth orbit weight makes the total annular frequency range*

$$|r| > X^\omega Hq/J \quad (65)$$

contribute $O_{A,\mathscr{D}}(X^{-A}Q^2)$ after all row, divisor, fixed-residue, and Mellin integrations.

Proof. Work first with one residue cell and one Mellin component. For a squarefree pair (u, v) in the annulus, put $q = [u, v]$, and let $Z_{u,v;\tau}(\xi)$ be its unweighted row fiber on the joint unit class $\xi \bmod q$. Thus

$$\sum_{\xi \in \mathcal{U}_q} |Z_{u,v;\tau}(\xi)| \leq \|x_\tau\|_1 \|y_\tau\|_1, \quad (66)$$

because the bounded row-pair mask only deletes or attenuates terms. After the fixed H_0 -cell is imposed, the Poisson formula has the same shape as the primitive formula, with its fixed modulus changed by an $O_h(1)$ factor. We use the normalization Hq below; replacing H by that fixed modulus changes only the implied constant.

The absolute tail functional for this Mellin component is bounded by

$$\begin{aligned} \mathcal{T}_\omega^{\text{ann}}(\tau) := & \sum_{\substack{u,v \leq T \\ \max\{u,v\} > S}} |b_R(u)b_R(v)| \frac{J}{Hq} \sum_{|r| > X^\omega Hq/J} |c_H(r)| \left| \widehat{\Phi}_\tau \left(\frac{rJ}{Hq} \right) \right| \\ & \times \sum_{\xi \in \mathcal{U}_q} |Z_{u,v;\tau}(\xi)|. \end{aligned} \quad (67)$$

This pairwise majorant also dominates a tail formed after lcm or conductor aggregation, by the triangle inequality. For every $N \geq 2$, the Fourier transforms in the physical smooth family satisfy

$$|\widehat{\Phi}_\tau(y)| \ll_N (1 + |\tau|)^N (1 + |y|)^{-N}.$$

Write $a_q = J/(Hq)$. An integral comparison, with the first lattice point treated separately when $X^\omega/a_q < 1$, gives uniformly in q

$$a_q \sum_{|r| > X^\omega/a_q} (1 + a_q|r|)^{-N} \ll_N X^{-\omega(N-1)}. \quad (68)$$

This is the required cancellation of the Poisson prefactor $J/(Hq)$ against the frequency-lattice spacing; no factor q/J remains. Since H is fixed, $|c_H(r)| \ll_H 1$. Moreover, the fixed-power support and the soft divisor-coefficient bound give a fixed constant C_{div} such that

$$\sum_{u,v \leq T} |b_R(u)b_R(v)| \ll X^{C_{\text{div}}}. \quad (69)$$

All lcm labels merely repartition these pairs, and the number and total variation of the fixed H_0 -residue cells are $O_h(1)$. Combining (66)–(69) yields

$$\mathcal{T}_\omega^{\text{ann}}(\tau) \ll_{N,\mathscr{D}} X^{C_{\text{div}} - \omega(N-1)} (1 + |\tau|)^N \|x_\tau\|_1 \|y_\tau\|_1. \quad (70)$$

Choose N so that $\omega(N-1) > A + C_{\text{div}}$, integrate against $d|\nu|$, and use (64) with the N -th weighted moment. Finally sum the $O_h(1)$ residue cells. This proves

$$\int_{\mathbb{R}^4} \mathcal{T}_\omega^{\text{ann}}(\tau) d|\nu|(\tau) \ll_{A,\mathscr{D}} X^{-A}Q^2,$$

as claimed. $\square$

5 The symmetric annular exact-conductor box

We treat every nonzero additive frequency of the raw annular joint lattice in (31). The lower annular restriction is retained in exact identities and may be discarded only when enlarging a nonnegative energy.

5.1 Joint Poisson and pairwise character coordinates

We fix all Fourier and finite-group normalizations in one place. Write

$$e(t) = e^{2\pi it}, \quad e_q(a) = e(a/q), \quad c_H(r) = \sum_{a \in (\mathbb{Z}/H\mathbb{Z})^\times} e_H(ra), \quad \widehat{F}(y) = \int_{\mathbb{R}} F(x) e(-xy) dx. \quad (71)$$

Here $F \in C_c^\infty(\mathbb{R})$ is the common orbit weight of one separated Mellin component; in particular it is the same function for every divisor pair in that component. Put

$$\mathcal{U}_q = (\mathbb{Z}/q\mathbb{Z})^\times, \quad \mathfrak{X}(q) = \widehat{\mathcal{U}_q}. \quad (72)$$

Target primitivity gives $(H, q) = 1$, and throughout this section $\overline{H}$ denotes the inverse of H modulo q . For a function $G : \mathcal{U}_q \rightarrow \mathbb{C}$, our multiplicative Fourier convention is

$$\widehat{G}^\times(\chi) = \sum_{\xi \in \mathcal{U}_q} G(\xi) \overline{\chi(\xi)}, \quad G(\xi) = \frac{1}{\varphi(q)} \sum_{\chi \in \mathfrak{X}(q)} \widehat{G}^\times(\chi) \chi(\xi). \quad (73)$$

For one effective squarefree divisor pair write

$$g = (u, v), \quad u = gc, \quad v = gd, \quad q = [u, v] = gcd. \quad (74)$$

Then g, c, d are pairwise coprime. The two target congruences are compatible exactly when

$$g \mid m - n. \quad (75)$$

When compatible, they determine one unit residue class $\rho_{m,n}^{u,v} \bmod q$. Suppressing only the fixed H_0 residue splitting, direct Poisson summation gives

$$\begin{aligned} & \sum_{\substack{j \in \mathbb{Z} \\ (j, H) = 1}} F(j/J) \mathbf{1}_{u|mj+h} \mathbf{1}_{v|nj+h} \\ &= \frac{J}{Hq} \sum_{r \in \mathbb{Z}} c_H(r) e_q(r \overline{H} \rho_{m,n}^{u,v}) \widehat{F}\left(\frac{rJ}{Hq}\right). \end{aligned} \quad (76)$$

The fixed residue cells change only the bounded local factor and a fixed modulus; they do not change any exponent below. Frequency zero was treated in section 4. We now assume $r \neq 0$.

Since $q = ud$ and $(u, d) = 1$, there is an exact product decomposition

$$\mathfrak{X}(q) \simeq \mathfrak{X}(u) \times \mathfrak{X}(d), \quad \chi \mapsto (\alpha, \gamma), \quad \text{cond } \chi = (\text{cond } \alpha)(\text{cond } \gamma). \quad (77)$$

For separated row weights x_m, y_n , put

$$c_{m,n}^{(g)} = x_m y_n \mathbf{m}(m, n) \mathbf{1}_{g|m-n}. \quad (78)$$

For this fixed pair, define the unweighted unit fiber

$$Z_{u,v}(\xi) = \sum_{\substack{m,n \\ g|m-n \\ \rho_{m,n}^{u,v} \equiv \xi \pmod{q}}} x_m y_n \mathbf{m}(m,n), \quad \xi \in \mathcal{U}_q, \quad (79)$$

with all primitive row restrictions retained. Its unit-fiber mean and discrepancy are

$$\overline{Z}_{u,v} = \frac{1}{\varphi(q)} \sum_{\xi \in \mathcal{U}_q} Z_{u,v}(\xi), \quad D_{u,v}(\xi) = Z_{u,v}(\xi) - \overline{Z}_{u,v}. \quad (80)$$

The exact multiplicative transform of the pair row fiber is

$$\widehat{Z}_{u,v}^\times(\chi) = \overline{\chi(-h)} \sum_{m,n} c_{m,n}^{(g)} \alpha(m) \gamma(n). \quad (81)$$

There is no residual $1/\varphi(g)$, and the second row character is not conjugated. This is finite CRT algebra; see TPC-22–24 [7–9].

For a dyadic conductor scale $F_0 \geq 1$, let

$$\mathfrak{X}_{F_0}(w) = \{\chi \in \mathfrak{X}(w) : \text{cond } \chi \asymp F_0\}, \quad (82)$$

where $F_0 = 1$ means the unique principal character. Under the CRT identification in (77), define the pairwise exact-conductor projection by

$$D_{u,v}^{F_1, F_2}(\xi) := \frac{1}{\varphi(q)} \sum_{\alpha \in \mathfrak{X}_{F_1}(u)} \sum_{\gamma \in \mathfrak{X}_{F_2}(d)} \widehat{D}_{u,v}^\times(\alpha \otimes \gamma)(\alpha \otimes \gamma)(\xi). \quad (83)$$

For every nonprincipal pair $(\alpha, \gamma) \neq (\mathbf{1}, \mathbf{1})$, centering changes no transform:

$$\widehat{D}_{u,v}^\times(\alpha \otimes \gamma) = \widehat{Z}_{u,v}^\times(\alpha \otimes \gamma). \quad (84)$$

Lemma 5.1 (Annular compatibility energy). *For every separated Mellin component from lemma 4.3,*

$$\boxed{\sum_{g \leq T} \sum_{m,n} |c_{m,n}^{(g)}|^2 \ll_\varepsilon X^\varepsilon \|x\|_2^2 \|y\|_2^2 \ll_\varepsilon Q^2 X^\varepsilon.} \quad (85)$$

Proof. The mask removes $m = n$. Therefore

$$\sum_{g \leq T} \sum_{m,n} |c_{m,n}^{(g)}|^2 \leq \sum_{m \neq n} |x_m|^2 |y_n|^2 \sum_{g|m-n} 1.$$

The last multiplicity is $X^{o(1)}$, independently of T , and the two row energies follow from (21) after absorbing fixed logarithms into X^ε . $\square$

5.2 Tensor energy at the true conductors

We use the exact-conductor large sieve in the normalization

$$\sum_{w \succ W} \frac{1}{\varphi(w)} \sum_{\substack{\chi \in \mathfrak{X}(w) \\ \text{cond } \chi \asymp F}} \left| \sum_r a_r \chi(r) \right|^2 \ll_\varepsilon X^\varepsilon \left(F + \frac{N}{F} \right) \sum_r |a_r|^2, \quad (86)$$

where a_r lies in an interval of length N . It follows from the primitive multiplicative large sieve at the true conductor after summing the inducing quotient [1, 7]. Put

$$\mathcal{K}_Q(F) = F + \frac{Q}{F}. \quad (87)$$

For dyadic Y, F_1, F_2 , let $\mathcal{E}_{Y;S,T}^{\text{ann}}(F_1, F_2)$ be the sum

$$\begin{aligned} & \sum_{\substack{g,c,d \text{ pairwise coprime} \\ gc, gd \leq T, \max\{gc, gd\} > S \\ g \mid cd \asymp Y}} \frac{|b_R(gc)b_R(gd)|^2}{\varphi(gcd)} \\ & \times \sum_{\alpha \in \mathfrak{X}_{F_1}(gc)} \sum_{\gamma \in \mathfrak{X}_{F_2}(d)} \left| \sum_{m,n} c_{m,n}^{(g)} \alpha(m) \gamma(n) \right|^2, \end{aligned} \quad (88)$$

with every primitive row restriction retained.

Theorem 5.2 (Symmetric annular conductor energy). *Uniformly for $Y \leq T^2$ and $F_1, F_2 \leq T$,*

$$\boxed{\mathcal{E}_{Y;S,T}^{\text{ann}}(F_1, F_2) \ll_{\varepsilon} X^{\varepsilon} \|x\|_2^2 \|y\|_2^2 \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2) \ll_{\varepsilon} Q^2 X^{\varepsilon} \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2).} \quad (89)$$

Proof. Fix g , split the two ambient moduli dyadically, and enlarge the nonnegative sum by deleting the annular lower bound, divisibility, and coprimality restrictions after the dyadic ranges have been retained. Apply (86) in the m -coordinate in its Hilbert-valued form, leaving the n -coordinate as a vector, and then apply it in the n -coordinate. Since both row variables occupy intervals of length $O(Q)$, the result for fixed g is

$$\ll_{\varepsilon} X^{\varepsilon} \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2) \sum_{m,n} |c_{m,n}^{(g)}|^2.$$

Sum over g and invoke lemma 5.1. The soft divisor coefficients and logarithmically many subdivisions are absorbed by X^{ε} . $\square$

5.3 Two estimates for one scalar cell

Assume the inherited no-wrap margin

$$J \geq 4HX^{\varepsilon_0} \quad (90)$$

for a fixed $\varepsilon_0 > 0$, and choose once and for all

$$0 < \omega < \varepsilon_0. \quad (91)$$

Restricting (76) to

$$0 < |r| \leq X^{\omega} Hq/J \quad (92)$$

keeps $|r| < q/4$ for large X . For the fixed divisor pair $u = gc, v = gd$, define the retained exact-conductor scalar

$$\mathcal{L}_{u,v}^{F_1, F_2, \leq \omega} := \frac{J}{Hq} \sum_{0 < |r| \leq X^{\omega} Hq/J} c_H(r) \hat{F}\left(\frac{rJ}{Hq}\right)$$

$$\times \sum_{\xi \in \mathcal{U}_q} D_{u,v}^{F_1, F_2}(\xi) e_q(r \overline{H} \xi). \quad (93)$$

The scalar annular cell is the pairwise absolute aggregate

$$\mathfrak{L}_{Y;S,T}^{\text{ann}}(F_1, F_2) := \sum_{\substack{g,c,d \text{ pairwise coprime} \\ gc, gd \leq T, \max\{gc, gd\} > S \\ g \mid c \mid d \asymp Y}} |b_R(gc) b_R(gd)| |\mathcal{L}_{gc, gd}^{F_1, F_2, \leq \omega}|. \quad (94)$$

Thus $\mathfrak{L}^{\text{ann}}$ is a sum of projections formed *inside each genuine unit fiber*; it is not the norm of a fictitious global rectangular projection.

Lemma 5.3 (Homogeneous restricted-frequency norm route). *For every exact-conductor cell,*

$$\mathfrak{L}_{Y;S,T}^{\text{ann}}(F_1, F_2) \ll_{\varepsilon} X^{\varepsilon} \|x\|_2 \|y\|_2 \sqrt{JY \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)} \ll_{\varepsilon} Q X^{\varepsilon} \sqrt{JY \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)}. \quad (95)$$

Proof. For one q , set

$$\mathcal{P}_q(\xi) = \frac{J}{Hq} \sum_{0 < |r| \leq X^{\omega} Hq/J} c_H(r) \widehat{F}\left(\frac{rJ}{Hq}\right) e_q(r \overline{H} \xi).$$

Multiplicative Parseval gives the exact block identity

$$\|D_{u,v}^{F_1, F_2}\|_{\ell^2(\mathcal{U}_q)}^2 = \frac{1}{\varphi(q)} \sum_{\substack{\alpha \in \mathfrak{X}_{F_1}(u) \\ \gamma \in \mathfrak{X}_{F_2}(d)}} |\widehat{D}_{u,v}^{\times}(\alpha \otimes \gamma)|^2. \quad (96)$$

The no-wrap inequality makes distinct retained r 's incongruent modulo q . Additive orthogonality on all residues, followed by Schwartz decay of $\widehat{F}$, therefore gives

$$\begin{aligned} \sum_{\xi \in \mathcal{U}_q} |\mathcal{P}_q(\xi)|^2 &\leq \sum_{\xi \bmod q} |\mathcal{P}_q(\xi)|^2 \\ &= \frac{J^2}{H^2 q} \sum_{0 < |r| \leq X^{\omega} Hq/J} |c_H(r)|^2 \left| \widehat{F}\left(\frac{rJ}{Hq}\right) \right|^2 \ll_{F,H} J. \end{aligned} \quad (97)$$

Multiplicative Parseval and Cauchy–Schwarz now imply

$$\begin{aligned} |\mathcal{L}_{u,v}^{F_1, F_2, \leq \omega}| &\leq \|\mathcal{P}_q\|_{\ell^2(\mathcal{U}_q)} \|D_{u,v}^{F_1, F_2}\|_{\ell^2(\mathcal{U}_q)} \\ &\ll_{F,H} \left\{ \frac{J}{\varphi(q)} \sum_{\substack{\alpha \in \mathfrak{X}_{F_1}(u) \\ \gamma \in \mathfrak{X}_{F_2}(d)}} |\widehat{D}_{u,v}^{\times}(\alpha \otimes \gamma)|^2 \right\}^{1/2}. \end{aligned} \quad (98)$$

There are $O_{\varepsilon}(YX^{\varepsilon})$ squarefree triples g, c, d with $gcd \asymp Y$. Sum (98) over those triples and apply Cauchy–Schwarz across divisor pairs. By (84), the resulting squared factor is bounded by $\mathfrak{L}_{Y;S,T}^{\text{ann}}(F_1, F_2)$, while the counting factor is $O_{\varepsilon}(JYX^{\varepsilon})$. The homogeneous estimate in [theorem 5.2](#) proves the first inequality in (95); (21) proves the second. $\square$

There is also an exact-conductor induced-Gauss route. If $f_1 = \text{cond } \alpha$ and $f_2 = \text{cond } \gamma$, put $f = f_1 f_2$ and $e = q/f$. Squarefreeness and the product decomposition imply $(e, f) = 1$, and finite CRT algebra gives

$$\left| \sum_{\xi \in \mathcal{U}_q} (\alpha \otimes \gamma)(\xi) e_q(r \overline{H} \xi) \right| \leq (f_1 f_2)^{1/2} |c_e(r)|. \quad (99)$$

The two primitive Gauss factors produce $(f_1 f_2)^{1/2}$; the remaining inducing quotient produces the Ramanujan sum. At nonzero frequency,

$$\sum_{r \neq 0} |c_H(r) c_e(r)| \left| \widehat{F} \left(\frac{rJ}{Hq} \right) \right| \ll_{F,H} \frac{Hq}{J} \tau(e). \quad (100)$$

The exclusion $r \neq 0$ is essential here: $c_e(0) = \varphi(e)$, and that frequency belongs to the physical zero theorem.

Lemma 5.4 (Homogeneous induced-Gauss route). *For every nonprincipal exact-conductor cell,*

$$\mathfrak{L}_{Y;S,T}^{\text{ann}}(F_1, F_2) \ll_{\varepsilon} X^{\varepsilon} \|x\|_2 \|y\|_2 F_1 F_2 \sqrt{\mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)} \ll_{\varepsilon} Q F_1 F_2 X^{\varepsilon} \sqrt{\mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)}. \quad (101)$$

Proof. Insert (83) into (93) before summing r . For fixed u, d , squarefreeness gives the elementary character count

$$\#\mathfrak{X}_{F_i}(w) \ll_{\varepsilon} F_i X^{\varepsilon} \quad (w \text{ squarefree}). \quad (102)$$

Indeed, there are X^{ε} possible conductor divisors of w , and each supports $O(F_i)$ primitive characters. Apply (99) and (100); then Cauchy–Schwarz in the characters gives, for one divisor pair,

$$|\mathcal{L}_{u,v}^{F_1, F_2, \leq \omega}| \ll_{\varepsilon, F, H} \frac{F_1 F_2 X^{\varepsilon}}{\varphi(q)} \left\{ \sum_{\substack{\alpha \in \mathfrak{X}_{F_1}(u) \\ \gamma \in \mathfrak{X}_{F_2}(d)}} |\widehat{D}_{u,v}^{\times}(\alpha \otimes \gamma)|^2 \right\}^{1/2}. \quad (103)$$

Restricting the frequency sum to the retained range only decreases this absolute majorant. Sum (103) over the annular divisor pairs and use Cauchy–Schwarz with weights

$$\frac{|b_R(u) b_R(v)|^2}{\varphi(q)} \quad \text{and} \quad \frac{1}{\varphi(q)}.$$

The first factor is the square root of the conductor energy, and

$$\sum_{\substack{g, c, d \text{ pairwise coprime} \\ g c d \asymp Y}} \frac{1}{\varphi(g c d)} \ll_{\varepsilon} X^{\varepsilon}. \quad (104)$$

Indeed, after grouping by $q = g c d$, the number of pairwise coprime factorizations is at most $3^{\omega(q)}$, while $q/\varphi(q) \ll_{\varepsilon} X^{\varepsilon}$; summing $1/q$ over $q \asymp Y$ costs $O(1)$.

The homogeneous form of [theorem 5.2](#) proves the first inequality in (101); the physical row energy proves the second. $\square$

Both lemmas estimate the same pairwise scalar cell (94). We may therefore take their minimum:

Theorem 5.5 (Symmetric annular cell bound). *For every nonprincipal conductor cell,*

$$\boxed{\mathfrak{L}_{Y;S,T}^{\text{ann}}(F_1, F_2) \ll_{\varepsilon} Q X^{\varepsilon} \sqrt{\mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)} \min\{\sqrt{JY}, F_1 F_2\}.} \quad (105)$$

The phrase “nonprincipal” excludes only the global cell $(F_1, F_2) = (1, 1)$. Cells with one principal coordinate remain in (105).

5.4 Exact nonprincipal reassembly

For a fixed denominator q , assemble the annular unit fiber before taking any conductor projection:

$$Z_{q;S,T}^{\text{ann}}(\xi) := \sum_{\substack{u,v \leq T, \max\{u,v\} > S \\ [u,v]=q}} b_R(u) b_R(v) Z_{u,v}(\xi), \quad \xi \in \mathcal{U}_q. \quad (106)$$

Put

$$\overline{Z}_{q;S,T}^{\text{ann}} = \frac{1}{\varphi(q)} \sum_{\xi \in \mathcal{U}_q} Z_{q;S,T}^{\text{ann}}(\xi), \quad D_{q;S,T}^{\text{ann}} = Z_{q;S,T}^{\text{ann}} - \overline{Z}_{q;S,T}^{\text{ann}}. \quad (107)$$

The full nonzero discrepancy packet at q is

$$\mathscr{J}_{q;S,T}^{\text{disc}, \neq 0} := \frac{J}{Hq} \sum_{r \neq 0} c_H(r) \widehat{F} \left(\frac{rJ}{Hq} \right) \sum_{\xi \in \mathcal{U}_q} D_{q;S,T}^{\text{ann}}(\xi) e_q(r \overline{H} \xi). \quad (108)$$

Let $\mathscr{J}_{q;S,T}^{\text{disc}, \neq 0, \leq \omega}$ denote the same expression with $0 < |r| \leq X^{\omega} Hq/J$. At a fixed Mellin component, linearity of (83) and the pairwise triangle inequality give the retained reassembly bound

$$\sum_{q \asymp Y} |\mathscr{J}_{q;S,T}^{\text{disc}, \neq 0, \leq \omega}| \leq \sum_{\substack{F_1, F_2 \leq T \\ (F_1, F_2) \neq (1, 1)}} \mathfrak{L}_{Y;S,T}^{\text{ann}}(F_1, F_2). \quad (109)$$

The Fourier tail is removed from this single undecomposed packet, before exact-conductor projection. Consequently, after restoring the fixed residue cells and integrating the Mellin representation from lemma 4.3, (109) and lemma 4.4 give, for each dyadic Y ,

$$\mathscr{J}_{q;S,T}^{\text{phys}, \text{disc}, \neq 0} := \sum_{\text{fixed residue cells}} \int_{\mathbb{R}^4} \mathscr{J}_{q;S,T,\tau}^{\text{disc}, \neq 0} d\nu(\tau).$$

Thus

$$\boxed{\sum_{q \asymp Y} |\mathscr{J}_{q;S,T}^{\text{phys}, \text{disc}, \neq 0}| \leq \int \sum_{\substack{F_1, F_2 \leq T \\ (F_1, F_2) \neq (1, 1)}} \mathfrak{L}_{Y;S,T,\tau}^{\text{ann}}(F_1, F_2) d|\nu|(\tau) + O_{A,\mathscr{D}}(X^{-A} Q^2).} \quad (110)$$

Here the conductor scales are dyadic, so the right side has only $O(\log^2(2T))$ cells. The error is precisely the physical tail in lemma 4.4; no pair-separated tail hypothesis is being inserted.

5.5 Principal mean and reassembly

For one divisor pair the principal multiplicative character produces the unit-fiber mean. In the notation above its assembled nonzero packet is

$$\mathcal{J}_{q;S,T}^{\text{mean},\neq 0} := \frac{J}{Hq} \sum_{r \neq 0} c_H(r) \widehat{F}\left(\frac{rJ}{Hq}\right) \overline{Z}_{q;S,T}^{\text{ann}} \sum_{\xi \in \mathcal{U}_q} e_q(r\overline{H}\xi). \quad (111)$$

Thus the last sum is the Ramanujan sum $c_q(r)$. Poisson summation over reduced classes and the elementary lcm cost

$$\sum_{u,v \leq T} \frac{1}{[u,v]} \ll \log^3(2T) \quad (112)$$

give the annular principal estimate

$$\sum_q |\mathcal{J}_{q;S,T}^{\text{mean},\neq 0}| \ll_{\varepsilon, \mathcal{D}} Q^2 X^\varepsilon. \quad (113)$$

It is power-saving relative to $XQ = Q^2 J$ whenever $1 - \beta > 0$ with fixed margin. The nonprincipal part reassembles according to (110), whose single undecomposed frequency-tail error is arbitrarily power-saving. No condition $T < J$ is used: the raw annular identity has already removed the two long single channels before this section begins.

6 The sharp symmetric conductor ledger

We now optimize the two estimates in [theorem 5.5](#) rather than retaining the earlier coarse low-low face.

6.1 Cell exponents

Write

$$Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad T = X^{t+o(1)}, \quad (114)$$

and, on one denominator and conductor cell,

$$Y = X^{y+o(1)}, \quad F_i = X^{f_i+o(1)}, \quad k_i = \max\{f_i, \beta - f_i\}. \quad (115)$$

The symmetric upper box gives

$$0 \leq y \leq 2t, \quad 0 \leq f_1, f_2 \leq t. \quad (116)$$

The natural scale XQ has exponent $1 + \beta$. The norm and Gauss routes in (105) have saving exponents

$$\eta_A(y, f_1, f_2) = \frac{\beta + 1 - y - k_1 - k_2}{2}, \quad \eta_B(f_1, f_2) = 1 - f_1 - f_2 - \frac{k_1 + k_2}{2}. \quad (117)$$

Decreasing y only improves η_A , so the relaxed-box minimax is

$$M_\beta(t) := \inf_{0 \leq f_1, f_2 \leq t} \max\{\eta_A(2t, f_1, f_2), \eta_B(f_1, f_2)\}. \quad (118)$$

Proposition 6.1 (Exact relaxed-box minimax). *For $1/2 \leq \beta < 1$,*

$$M_\beta(t) = \begin{cases} 1 - \beta - t, & 0 \leq t \leq (1 - \beta)/2, \\ \frac{3 - 3\beta - 2t}{4}, & (1 - \beta)/2 \leq t \leq \beta/2, \\ \frac{3 - \beta - 6t}{4}, & \beta/2 \leq t \leq (3\beta - 1)/2, \\ \frac{\beta + 1 - 4t}{2}, & t \geq (3\beta - 1)/2. \end{cases} \quad (119)$$

The formula is exact for the relaxed rectangle (116). It is therefore a valid lower bound for the occupied arithmetic cells, which form a subset of that rectangle.

Proof. Suppose first that both conductors are low: $f_i \leq \beta/2$. Put $\sigma = f_1 + f_2$. Then

$$\eta_A = \frac{1 - \beta - 2t + \sigma}{2}, \quad \eta_B = 1 - \beta - \frac{\sigma}{2}. \quad (120)$$

The two affine functions meet at

$$\sigma_* = t + \frac{1 - \beta}{2}, \quad (121)$$

where their common value is

$$\frac{3 - 3\beta - 2t}{4}. \quad (122)$$

If $0 \leq t \leq (1 - \beta)/2$, the meeting point lies beyond the available interval $0 \leq \sigma \leq 2t$; the minimizing endpoint $\sigma = 2t$ gives $1 - \beta - t$. If $(1 - \beta)/2 \leq t \leq \beta/2$, the meeting point is available and gives (122).

Now let $t \geq \beta/2$. In a high-high cell, both savings decrease as either conductor increases, so the regional minimizer is $f_1 = f_2 = t$. There

$$\eta_A = \frac{\beta + 1 - 4t}{2}, \quad \eta_B = 1 - 3t. \quad (123)$$

In the range relevant to the global transition, the first is the larger of the two.

For a mixed cell write

$$x = f_{\text{low}} \leq \frac{\beta}{2}, \quad z = f_{\text{high}} \geq \frac{\beta}{2}.$$

Increasing z decreases both savings, so $z = t$. At that edge,

$$\eta_A = \frac{1 - 3t + x}{2}, \quad \eta_B = 1 - \frac{\beta}{2} - \frac{3t}{2} - \frac{x}{2}. \quad (124)$$

They meet at

$$x_* = \frac{1 - \beta}{2}, \quad \eta_A = \eta_B = \frac{3 - \beta - 6t}{4}. \quad (125)$$

The witness is admissible because $\beta \geq 1/2$. The low-low regional value is never smaller than the mixed value when $t \geq \beta/2$: their difference is $t - \beta/2$ whenever the interior low witness is available, and the relevant boundary case is larger as well. Thus the global minimum is the smaller of

$$\frac{3 - \beta - 6t}{4} \quad \text{and} \quad \frac{\beta + 1 - 4t}{2}.$$

These cross at $t = (3\beta - 1)/2$, proving the last two branches. Continuity at all three boundaries is immediate. $\square$

Remark 6.2 (What “sharp” means). The proposition exactly optimizes the two available conductor bounds over the relaxed box. It does not assert that every equality witness is occupied by an arithmetic divisor cell, nor that no stronger third estimate exists. Therefore a zero or negative value marks the boundary of this two-route method, not an arithmetic impossibility theorem.

6.2 Beyond the Poisson face

The upper cutoff lies beyond the old Poisson length when

$$t > 1 - \beta. \tag{126}$$

In that region [proposition 6.1](#) becomes

$$M_\beta(t) = \begin{cases} \frac{3(1 - \beta) - 2t}{4}, & 1 - \beta < t \leq \beta/2, \\ \min \left\{ \frac{\beta + 1 - 4t}{2}, \frac{3 - \beta - 6t}{4} \right\}, & t \geq \beta/2. \end{cases} \tag{127}$$

The low-conductor branch is positive in

$$\boxed{\frac{2}{3} < \beta < 1, \quad 1 - \beta < t < \min \left\{ \frac{\beta}{2}, \frac{3(1 - \beta)}{2} \right\}.} \tag{128}$$

The high/mixed branch is positive in

$$\boxed{\frac{3}{5} < \beta < \frac{3}{4}, \quad \max \left\{ 1 - \beta, \frac{\beta}{2} \right\} < t < \min \left\{ \frac{\beta + 1}{4}, \frac{3 - \beta}{6} \right\}.} \tag{129}$$

Their union is nonempty for every

$$\boxed{\frac{3}{5} < \beta < 1.} \tag{130}$$

The old face $\beta < 2/3$ is therefore not intrinsic to the two cell estimates in the annular joint problem. Other parts of the residual program may still impose it.

For a strict passage from below, one additionally requires

$$s < 1 - \beta < t. \tag{131}$$

With $S = R = X^{1/2 - \delta + o(1)}$ and $T = RX^{\xi + o(1)}$, a strict high/mixed sufficient window is

$$\boxed{\begin{aligned} & \frac{3}{5} < \beta < \frac{3}{4}, \quad \frac{1}{2} - \delta < 1 - \beta, \\ & \max \left\{ \frac{1}{2} + \delta - \beta, \frac{\beta - 1 + 2\delta}{2} \right\} < \xi < \min \left\{ \frac{\beta + 4\delta - 1}{4}, \frac{6\delta - \beta}{6} \right\}. \end{aligned}} \tag{132}$$

The first lower bound makes $T > J$, the second makes $t > \beta/2$, and the separate condition on δ makes $S < J$. Omitting the branch condition $t \geq \beta/2$ would make [\(132\)](#) false outside the high/mixed regime.

All inequalities in this section are strict and must be used with a fixed margin.

6.3 Two exact rational samples

The first sample is

$$\delta = \frac{3}{20}, \quad \beta = \frac{31}{50}, \quad s = \frac{7}{20}, \quad t = \frac{39}{100}. \quad (133)$$

Then

$$S = X^{7/20} < J = X^{19/50} < T = X^{39/100}, \quad (134)$$

and

$$\frac{\beta + 1 - 4t}{2} = \frac{3}{100}, \quad \frac{3 - \beta - 6t}{4} = \frac{1}{100}. \quad (135)$$

Thus

$$\boxed{M_\beta(t) = \frac{1}{100}.} \quad (136)$$

This is a genuine strict passage sample.

The second sample tests the sharp improvement against the published Maynard source profile used in TPC-18. Put

$$\sigma = \frac{1}{10000}, \quad \lambda = \frac{10}{21} - \sigma = \frac{99979}{210000}, \quad \delta = \frac{7}{60}, \quad (137)$$

and choose

$$\beta = \frac{267}{400}, \quad s = \frac{23}{60}, \quad t = \frac{193}{500}. \quad (138)$$

The opened divisor exponent and its available endpoint are

$$d_* = \beta - \lambda = \frac{10049}{52500}, \quad v_* = \frac{1}{4} - \frac{\delta}{2} = \frac{23}{120}, \quad (139)$$

with exact margin

$$v_* - d_* = \frac{9}{35000} > 0. \quad (140)$$

Moreover,

$$\beta - \frac{2}{3} = \frac{1}{1200}, \quad t - s = \frac{1}{375}, \quad \lambda - t = \frac{18919}{210000} > 0. \quad (141)$$

The two sharp conductor faces are

$$\frac{\beta + 1 - 4t}{2} = \frac{247}{4000}, \quad \frac{3 - \beta - 6t}{4} = \frac{33}{8000}, \quad (142)$$

so

$$\boxed{M_\beta(t) = \frac{33}{8000}.} \quad (143)$$

Indeed,

$$\frac{1}{42} + 3\sigma < \delta < \frac{1}{4}, \quad D_0 = X^{1/21-\sigma} < D \leq V = X^{v_*}, \quad K = X^{1-\lambda} > 2V,$$

with fixed exponent margins, and $J = K/D$ has exponent $(1 - \lambda) - d_* = 1 - \beta$. Thus the sample is compatible with that source profile and even satisfies $L > 2T$. However,

$$J = X^{133/400} < S < T, \quad (144)$$

so it proves annular cutoff stability beyond the Poisson face, not a strict passage from an independently closed calibrated base square.

7 Physical cutoff stability and truncated-square closure

We now combine the exact annular identity, physical zero theorem, and sharp nonzero conductor ledger.

7.1 The nonzero physical packet

Let

$$(\mathcal{Q}_T^{\text{phys}} - \mathcal{Q}_S^{\text{phys}})^{\neq 0}$$

denote the sum of all nonzero additive frequencies of the raw annular joint lattice in (31), after the two drift terms have been set aside.

Theorem 7.1 (Nonzero annular conductor theorem). *Assume the primitive physical interface of section 2, the divisor-independent off-diagonal mask, the no-wrap margin (90), and fixed-power support $T \leq X^{C_T}$. Write*

$$Q = X^{\beta+o(1)}, \quad T = X^{t+o(1)}, \quad \frac{1}{2} \leq \beta < 1,$$

with $1 - \beta$ bounded below by a fixed positive margin, and suppose

$$M_\beta(t) > 0 \quad (145)$$

with a fixed margin. Then there is a fixed $\eta > 0$ such that

$$|(\mathcal{Q}_T^{\text{phys}} - \mathcal{Q}_S^{\text{phys}})^{\neq 0}| \ll_{\mathscr{D}} XQX^{-\eta}. \quad (146)$$

One may take any sufficiently small fixed η below the conductor margin $M_\beta(t)$ and the principal-mean margin $1 - \beta$, after the auxiliary X^ε losses are chosen.

Proof. For one Mellin component, decompose the nonprincipal spectrum into dyadic denominator and coordinate-conductor cells. Apply theorem 5.5 to each cell and proposition 6.1 to its exponents. The fixed positive margin absorbs the logarithmically many cells and the row-energy factor in (63). The global principal cell is (113), which is below $XQ = Q^2J$ by the fixed power $J = X^{1-\beta+o(1)}$. The physical Fourier tail is arbitrarily power-saving by lemma 4.4. Finally, integrate the Mellin components using the weighted moments in lemma 4.3. The fixed H_0 -residue cells change only the implied constant. $\square$

7.2 Main cutoff theorem

We use the fixed-data convention $\mathscr{D}$ introduced in section 2.

Theorem 7.2 (Physical annular cutoff stability). *Retain the hypotheses of [theorem 7.1](#). Assume*

$$S = X^{s+o(1)}, \quad s > 0, \quad M_\beta(t) > 0 \quad (147)$$

with fixed margins. Then, for every fixed $B > 0$,

$$\boxed{\mathcal{Q}_T^{\text{phys}} - \mathcal{Q}_S^{\text{phys}} \ll_{B, \mathcal{D}} XQ(\log X)^{-B}.} \quad (148)$$

The same theorem applies to a finite bounded-overlap reassembly of physical dyadic, content, sign, and fixed-residue cells; its fixed polylogarithmic cost is absorbed by choosing the zero-mode calibration exponent afterward.

Proof. Use [proposition 2.1](#). The nonzero raw annular joint term has a fixed-power saving by [theorem 7.1](#). The physical zero contribution together with the two separately bounded full drifts has arbitrary logarithmic saving by [corollary 4.2](#). A fixed-power saving is smaller than any prescribed negative power of $\log X$ for sufficiently large X , which proves (148). Finite cell reassembly costs only a fixed power of $\log X$. $\square$

Remark 7.3 (What the theorem closes). The theorem controls a cutoff difference. It does not estimate $\mathcal{Q}_S^{\text{phys}}$ and therefore does not by itself estimate $\mathcal{Q}_T^{\text{phys}}$. This distinction is essential in the high- β sample ([138](#)), where both S and T exceed J .

Corollary 7.4 (Selected truncated-square closure). *In addition to the hypotheses of [theorem 7.2](#), suppose the same physical S -cutoff packet satisfies*

$$\mathcal{Q}_S^{\text{phys}} \ll_{B, \mathcal{D}} XQ(\log X)^{-B} \quad (149)$$

for every fixed $B > 0$. Then

$$\boxed{\mathcal{Q}_T^{\text{phys}} \ll_{B, \mathcal{D}} XQ(\log X)^{-B}.} \quad (150)$$

The estimate (149) is an independent calibrated base input. TPC-22–23 provide centered selected-square estimates, but the scalar terms required to pass from those centered packets to the present $P_{m,S}P_{n,S}$ packet are not proved in this paper. Accordingly, this corollary is conditional even in the strict sample ([133](#)).

Proof. Apply the triangle inequality to

$$\mathcal{Q}_T^{\text{phys}} = \mathcal{Q}_S^{\text{phys}} + (\mathcal{Q}_T^{\text{phys}} - \mathcal{Q}_S^{\text{phys}})$$

and use (149) and (148). $\square$

Corollary 7.5 (Vanishing physical zero functional). *If*

$$\vartheta_{H_0}(\alpha, \gamma) \widehat{\mathcal{W}}_{\alpha, \gamma}(0) = 0$$

for every retained row pair and residue cell, then the raw annular physical zero term vanishes. The separately bounded full drifts remain. Under the same conductor hypotheses, some fixed $\eta' > 0$ satisfies

$$\mathcal{Q}_T^{\text{phys}} - \mathcal{Q}_S^{\text{phys}} \ll XQX^{-\eta'}. \quad (151)$$

This last statement is optional extra cancellation. The main theorem does not assume it.

7.3 The next analytic wall

Let $U \asymp X$ contain the complete target support. The theorem does not estimate

$$L_{m;T,U} = \sum_{\substack{T < u \leq U \\ u \mid mj+h}} a(u). \quad (152)$$

TPC-23 reflects a divisor $u > T$ to a complement $r < (mj + h)/T$. For fixed complement divisors r, s , the two quotient variables are affine forms whose determinant is

$$\frac{h(m-n)}{(r,s)} \neq 0. \quad (153)$$

The remaining coefficient is a two-point product of Möbius signs. Taking absolute values already has natural-scale mass on the unit complement fiber, while the known logarithmically averaged fixed-form theorem has neither the required polynomial uniformity nor the sharp dyadic averaging [2, 8]. Continuing beyond T therefore requires a uniform affine-Möbius dispersion theorem, a sparse incidence theorem, or a new exact recombination. It is not a formal consequence of [theorem 7.2](#).

8 Exact certificate, logical status, and conclusion

The analytic estimates in this paper are symbolic. A deterministic finite certificate is included to audit the exact annular algebra, gcd normalization, and rational ledger identities most vulnerable to sign or boundary errors.

8.1 Certificate

The standard-library script

`experiments/tpc26_certificate.py`

uses `fractions.Fraction` and formal prime-log vectors. It checks:

- (1) the exact identity $P_T P_T - P_S P_S$ against the raw annular joint lattice and both negative drift terms on finite symbolic prefix data;
- (2) the partition of the raw annulus into the two mixed rectangles and the both-new square;
- (3) the direct finite both-new zero kernel against the (g, c, d) normal form in [lemma 3.2](#);
- (4) the coefficient identity (46) as an equality of formal logarithm vectors;
- (5) CRT compatibility and the joint zero density $[u, v]^{-1} \mathbf{1}_{(u,v) \mid m-n}$ on finite primitive examples;
- (6) the sharp minimax regional witnesses and boundary continuity on an exact rational grid;
- (7) the strict 1/100 passage sample; and
- (8) every source, support, row, and conductor margin in the $\beta > 2/3$ sample, including the saving 33/8000.

Independent ordinary and `python -O` runs are required to perform the same exact checks and generate byte-identical JSON. The repository README records the final JSON, source, and certificate hashes after the source audit. The archived JSON carries false flags for full

residual closure, positivity, a Hardy–Littlewood asymptotic, twin primes, and a general parity breakthrough.

The certificate does not test the classical zero-free region, the primitive multiplicative large sieve, any asymptotic Möbius cancellation, or the truth of a prime-pair conjecture. It is regression control for finite algebra and rational exponent arithmetic.

8.2 What is proved directly and what is inherited

The following parts are finite or elementary within the stated physical packet:

- (i) the calibrated prefix identity and exact annular recombination;
- (ii) the both-new gcd normal form;
- (iii) the division of the physical annular zero mode into mixed, transpose, both-new, and two single-density drift terms;
- (iv) the physical $O(Q^2)$ absolute row-pair mass once the TPC-25 provenance lemma is imported;
- (v) the symmetric pairwise character coordinates; and
- (vi) the exact relaxed-box minimax and the two rational samples.

The pointwise small- g zero estimate imports the uniform Möbius calibration proved in TPC-24 from the classical zero-free region. The nonzero spectrum imports the exact-conductor large-sieve mechanism of TPC-22–24. The smooth transfer and row norms import the physical provenance established in TPC-25. No conjectural Chowla or Elliott input is used in the cutoff theorem.

The role of the present paper is to prove that those inherited tools remain compatible after the three new support regions are first recombined into one annular joint object, and to optimize their symmetric conductor ledger exactly.

8.3 Status table

Object	Status after this paper
Both-new physical zero mode on $(S, T]^2$	Closed with arbitrary logarithmic saving by endpoint calibration, determinant divisibility, and actual $O(Q^2)$ row mass.
Raw annular joint nonzero spectrum	Closed with fixed-power saving whenever $M_\beta(t) > 0$.
Physical cutoff difference $Q_T^{\text{phys}} - Q_S^{\text{phys}}$	Closed with arbitrary logarithmic saving when $s > 0$, $M_\beta(t) > 0$, and all inherited interfaces hold with fixed margins.
Selected calibrated square through $T > J$	Closed only when the matching S -cutoff base estimate is also available.
High- β sample with $\beta > 2/3$	Annular cutoff stability proved inside actual opened-row geometry; no base-square closure imported.
Ultra-long $(T, U]$ complement	Not estimated; exact affine two-point Möbius fibers remain.
Full TPC-18 residual and endpoint reassembly	Not proved.
Hardy–Littlewood, positivity, twin primes, or general parity breach	Not claimed and not implied.

8.4 Conclusion

The both-new square is not blocked by the principal row mode found in TPC-25. That mode arose when a shell coefficient was frozen before its full endpoint structure was used. In the complete annulus the small- g main constants cancel, and the large- g layers are sparse divisors of the nonzero row determinant.

The successful order is

$$\begin{array}{l}
 \text{form the quadratic cutoff difference} \longrightarrow \text{recombine the entire raw annulus} \\
 \longrightarrow \text{extract physical zero} \longrightarrow \text{separate and resolve only the nonzero spectrum.}
 \end{array} \tag{154}$$

This closes a real missing component. Conditional on an independent matching calibrated base estimate, it pushes the whole selected truncated square through an upper cutoff $T > J$; that base estimate is not supplied here. The exact minimax also shows that the annular conductor mechanism itself survives beyond $\beta = 2/3$, though that fact must not be transferred automatically to the base square or the complete residual.

The next wall is qualitatively different. Divisor reflection in the ultra-long complement places two growing affine Möbius signs on the same short fiber. A future advance must obtain polynomially uniform two-point cancellation, exploit sparse incidence after summing the family, or find another exact recombination. Success would close one more analytic residual block; it would still not by itself produce a positive prime-pair main term.

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